



Performance assessment architecture for collaborative business processes in BPM-SOA-based environment



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ABSTRACT

To be competitive and flexible, companies engage in collaborations to develop and share their competences in order to cope with the dynamic environment. Collaborative business process evaluation helps to reflect the actual functioning of business process and their performance level. In this perspective, research in assessing collaborative business process performance presents relevant guidelines in order to adapt IT solutions when business requirements evolve. In this paper, we present an analysis and assessment approach for collaborative business processes in the service-oriented architecture in order to maintain their performance in competitive markets. Our approach proposes an evaluation method using execution traces of business process combined with a high-level assessment method using key performance indicators. Our main objectives are to track the execution of collaborative business process and to analyze the performance trajectory of a business process regarding the business performance level. To collect and structure the performance knowledge (execution and measurement), we create an ontological model-based knowledge repository in order to enrich the semantics of an evaluation business process. The precise track of execution data in our approach is able to identify events that disrupt the proper functioning of processes at the runtime. From an industrial case study, we can conclude that our ontological approach can target the performance assessment of collaborative business processes effectively.

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1. Introduction

Nowadays, companies need to be more competitive and agile in the global market. Therefore, they are investigating the building and maintaining of new collaborations in order to increase their productivity. Collaborative business processes (CBP) are becoming more indispensable and crucial to competitiveness in the dynamic environment (globalization, crisis, new competitors, etc.) in which organizations operate. Collaborative business processes describe how organizations operate, and therefore impact how organizations perform. In this perspective, research in enhancing business process management (BPM) capabilities (alignment business/information technology (IT), maturity, interoperability, etc.) presents relevant guidelines in order to adapt IT solutions when business requirements evolve.

Implementing collaborative process using service orientated architecture (SOA) helps enterprises to be more responsive and more flexible to an increasingly changing market environment. In addition, those organizations need to gain a better insight into the way their business processes (BP) are executed. The evaluation of business process execution illustrates various insights

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regarding the real execution of the BPs usage, the performance level, the reliability analysis, the QoS (quality of service) measures, etc.

In the literature, several existing proposals are concentrated on the business level performance and they focus mostly on indicators related to the business specifications, without considering the IT infrastructure which realizes their BPs. Other existing approaches also emphasize the assessment of IT level [33]. However, the business value of indicators related to IT infrastructure is an open question for researchers.

In order to address the above issues, we concentrate mainly on this research gap and we present in this article an approach for the analysis and assessment of the quality of CBPs. Our proposal includes an assessment model that integrates execution measures for BPs with services implementing them. We calculate technical indicators (TI) (execution duration, maturity, interoperability, etc.) from business process traces collected from the data resulting from IT infrastructure supporting BP. The main objective of these TIs is to evaluate the execution of BPs. Our approach also contains an ontological model-based knowledge repository to collect and structure the semantic of BP performance. We propose a set of key performance indicators (KPI) defined from the business specifications and collaborative objectives to evaluate the adequacy of management initiatives identified for the collaboration. Then, an aggregation model was developed. This model is a down-up model and it goes from the lowest level (applicative/execution level) to the highest level (business level). In order to project the TIs on the business level, we correlate two performance levels: business performance level and performance level aggregated from TI.

The rest of the article is organized as follows: in Section 2, we perform a literature review to collect existing proposals related to our research work. Section 3 contributes our methodology for the analysis and assessment of CBP in SOA taking into account the evolution of organizations events. Its main challenge is to maintain and improve CBP performance by analyzing, interoperability, maturity, etc. Section 4 presents our results and discussions. In Section 5, conclusions and future works are discussed.

2. Literature review

The work related to our proposition has three dimensions. First, we present a number of techniques in order to analyze business processes. Second, the research that deals with the assessment of business process models is discussed. Third, the semantic models which enrich the business process evaluation model are discussed.

2.1. Analysis of collaborative business process: process mining

Business process analysis (BPA) becomes very important to enable the successful improvement of business processes. BPA defines a set of methodologies, techniques and tools aimed at understanding and reasoning, from a process perspective, how an organization works, in order to detect problems and to find potential improvements [38]. The knowledge obtained from BPA presents an important starting point to enhance business processes.

Business activity monitoring (BAM) has been developed to extract knowledge from event logs to provide real-time information about the status and results of business processes [25]. In addition, business process intelligence (BPI) is an emerging field of research which is aimed at process monitoring and analysis for the process discovery, process efficiency, conformance checking, prediction and optimization of business processes [26].

Tracing and analyzing activities in information technologies (IT) is a field that appeared when the need to discover process models emerged [10]. A well-known technique, process mining [20] describes a posteriori analysis techniques exploiting the information recorded in data traces. According to [17], process mining currently has three possible applications: (1) the discovery of process knowledge; (2) conformance checking, i.e., detect, locate, explain and measure deviations between a defined model and the real process model mined from the available traces; and (3) enhancement and extension of a process model with the event data, e.g. enrich a process model with the performance information.

Process mining aims to fill the gap between activity traces obtained from the event logs (e.g. transaction logs in an ERP system or another DB) and defined process models. The main goal of the process mining technique is to extract knowledge from the historical log files and events [9]. This knowledge is used for process discovery, and for enhancing the quality of the process by detecting deviations and problems in the process model. A bottom-up process mining and simulation-based methodology are proposed in [57] in order to be employed for the redesign of business process. The methodology consists of defining important performance issues. A process model is “mined” and simulated as a representation of the existing situation, followed by the simulation of the redesigned process model as a prediction of a future situation. Then, the performance criteria of the current business process model and the redesigned business process model are compared. Collaborative process mining also needs special attention to ensure and meet the quality by all the stakeholders involved in the collaborative engineering of business processes.

There are various process mining algorithms that aim at discovering processes from the event logs, such as α -algorithm [28], hierarchical clustering [29], heuristic algorithm [30], genetic algorithm [19], etc. Many other tools emerged to support the process mining techniques, for example ProM [11], CPN tools [12], EMiT [13] and Disco [14]. The process mining tool ProM has been applied in many organizations ranging from municipalities and hospitals to high-tech manufacturers and multinationals. ProM provides several plug-ins to extract information from the real execution of the BP. By using this tool, Rebuge and Ferreira have presented a methodology based on the process mining for the analysis of healthcare processes. The event data collected was analyzed with a special-purpose tool as well as with the plug-ins available in the ProM framework [18]. The work of [56] contributes to a better understanding of collaboration processes that are distributed over various geographical locations. In this research, authors analyzed a large workflow process log with the ProM framework.

Several researchers investigated the difference and commonality between data mining and process mining. The objective of data mining is to extract knowledge from a huge quantity of data [47], while process mining aims to mine process models from data represented by the process execution workflows. Hence, the data is a base for both data mining and process mining [51]. However, the data is stored differently in these two cases. The data employed in data mining are often stored in databases, data warehouses, the World Wide Web or other information repository [47], while the data used in process mining are usually stored in logs of the system workflows. Clustering is a simple data mining technique, and does not lead to the extraction of a whole process model; nonetheless, clearly grouping trace can be a significant support for checking the conformance and verification of the semantic properties. For example, in [21], a framework has been proposed in order to support real-time adjustment and a posteriori analysis of business processes. This framework permits to retrieve traces of similar process executions. Moreover, it supports an automatic organization of the database traces through the application of hierarchical clustering techniques. Several researchers contributed different techniques for the cluster analysis to find groups in the data to explore data mining, such as discovering expressive process models by clustering log traces [22], context aware trace clustering [23], trace alignment in process mining [24], etc.

As a conclusion, the essence of process mining is the extraction of process knowledge from event logs recorded by information systems. In fact, we will use process mining that is identified as an approach with potential to reduce the time required for the process analysis, as well as to provide a more accurate picture of business processes which are more highly dynamic than traditional approaches for the process analysis.

2.2. Assessment of collaborative business process (top-down and down-up approaches)

Assessing the quality of a business process execution is a complex task. On the one hand, there are different perceptions of quality corresponding to different views (business, infrastructure IT, etc.). On the other hand, there are several approaches proposing diverse quality criteria, performance indicators and evaluation tools. Several existing proposals are mainly concentrated on the business level performance and focus mostly on the definition of KPIs related to the flow and business specifications, not explicitly taking into account the infrastructure which realizes and supports it. Guo et al. [7] constructed the BP evaluation indexes on four aspects: financial value, BP, customer's service and study innovation. They concentrated on indicators related to the business level, such as the product demand rate, product sale rate and production flexibility. Other scholars consider that performance includes two distinctive kinds of performance metrics regarding business indicators and IT indicators. For instance, Zeng et al. proposed the service-oriented business process (SOBP) [8]. The SOBP model comprises metrics relating to 3 levels: the business level (reliability, risk, time, cost and flexibility), the application level (response time of the system, system availability, system flexibility) and the IT infrastructure level (reliability of IT component, use of IT resources).

One of the most important factors to access the performance of CBP is *maturity* as it is important for the business permanence, improvement and sustainability of an organization. A maturity model helps organizations to assess the strengths and weaknesses of their business processes and make improvements. A business process maturity model (BPMM) assesses and improves a business process throughout its lifecycle by focusing on the necessary capabilities to perform [39]. Moreover, BPMMs aim to gradually increase business process performance [40]. In [41], the authors presented 150 available models addressing one or more components of BPM. Some models do not encompass all facets of BPM that are critical to progression. Others models are relevant to the management of a specific process and not to the management of all processes. BPMMs present a sequence of maturity levels and a step-by-step roadmap with goals and best practices to reach each consecutive maturity level [42]. For example, the OMG models focused on the business process optimization. For that, they considered five levels 'initial', 'managed', 'standardized', 'predictable' and 'innovating' [43]. Other BPMMs prefer emphasizing business process integration, such as McCormack and Johnson's levels of 'ad hoc', 'defined', 'linked' and 'integrated' [44]. We are investigating the capability maturity model integration (CMMI) model developed by Software Engineering Institute (SEI) [45] in our research study. It presents a framework that is based on the best practices for developing or improving processes and services that meet the business goals of an organization. Our CBPs are assessed using the CMMI model (initial, repeatable, defined, managed and optimized), which are elaborated in [46].

Another aspect for the evaluation of collaborative business processes is the ability to exchange information and data among the involved stakeholders of organizations. Collaborative business processes provide organizations a favorable context for the automated business process interoperability. Elmir and Bounabat [34] proposed a novel approach for assessing the interoperability of process-driven services by taking into account the three main aspects of interoperability: potentiality, compatibility and operational performance. They also developed a software tool that supports the proposed assessment method. The main objective of this tool is to track and control the evolution of interoperability degree over the time. Moreover, the proposed tool measures the required effort to reach a planned degree of interoperability.

Risk is one of the vital measures during the business process and can be defined as the probability of the deviation from the actual result of BP implementation and the expected target output. Kefan et al. proposed a holistic approach towards measuring risks of business process in the collaborative business processes [49]. They investigated risk in the business process comprised of risk involved in the operational environment of the business process, risk from the main parts of business process, and risk from the object of business process. Conforti et al. contribute real-time monitoring of the risk during all phases of the business processes in the executable business process models [50]. They designed sensors which interact with the monitoring component of a business process management system to prompt the results to process administrators who may take remedial actions.

The business value of IT has been widely discussed in the literature, but after several decades of research, organizations continue to look for ways to increase the value won from their IT investments [5]. Therefore, the assessment and measure of the added value by IT remains difficult and subject to diverse views [6]. In addition, they argued that the analysis of individual

capabilities of the organizational resources or IT assets do not produce challenging results in isolation, but rather to the emergent capabilities that arise from their combination. The existing approaches almost did not analyze the signification of IT on the business point of view. Therefore, we took this initiative to further explore this research aspect.

In recent years, semantic technologies have revived and unleashed a revolution of new capabilities in almost all domains of research. As a result, many of the latest contributions have enriched business process evaluation models with the semantic technologies to reap maximum benefits. Initial results showed that the role of the ontological approaches/models in the BP performance assessment is significant. This domain of research is highly related to our contribution presented in this paper which is elaborated in the next sections.

2.3. Semantic enrichment of business process evaluation models: ontological model

In recent years, there have been several researches regarding business process based on ontologies. Therefore, the most basic objective of an ontology is to formalize and conceptualize the domain knowledge. An ontology, as a formal specification, provides a framework for facilitating efficient knowledge-sharing in the collaborative business process development. Ontologies provide the semantic context by adding semantic information and define a set of terms and constraints that describe the structure and behavior of the business process. Many works emphasize the benefits and potential uses of ontologies in the field of business process analysis and assessment. Valiente et al. proposed an ITIL (information technology infrastructure library) based ontological approach focused on the implementation of processes in the context of information technology service management [1]. Their ontology provides a support for the executable service models with computational semantics. The proposed ontology provides a mechanism for managing interoperability, consistency checking and decision making. This approach defines the semantics and constraints associated with IT service, avoiding ambiguities and enabling a better and common understanding of IT service management processes. The main objective of Onto-ITIL is to overcome the gap existing between business needs and IT services delivered to customers. Pedrinaci et al. presented the core ontology for the business process analysis (COBRA), which is built on two ontologies referred to as *base ontology* and *time ontology* [2]. Base ontology provides definitions for the basic modelling concepts such as tasks, relations, functions, roles, numbers, etc. Time ontology provides a temporal reference by means of which one can determine temporal relations between elements. The objective of this model is to reduce gaps between semantics technologies and the ones currently deployed within BPM solutions. Brocke et al. defined the domain ontology as an ontology for linking processes and IT infrastructure (OLPIT), in order to model the relationship between IT resources and business processes for the purpose of measuring the business value of IT [3]. Pedrinaci and Domingue created process mining ontology (PMO) to capture events appearing during the lifecycle of business processes and combined it with additional mining information in order to support the analysis of enacted processes at different levels of abstraction from the fine grain technical details to coarse grain aspects at the business level [4].

All the approaches presented in this section address a specific facet of BP quality. Several existing proposals are mainly concentrated on the business level performance and focus mostly on the definition of KPIs related to business specifications, not explicitly taking into account the infrastructure which supports their BP. They don't emphasize the potential role of IT in the organization. Others consider that the performance contains two different types of performance indicators: the first type assesses the business level performance and the second type evaluates the IT level. In the framework of these approaches,

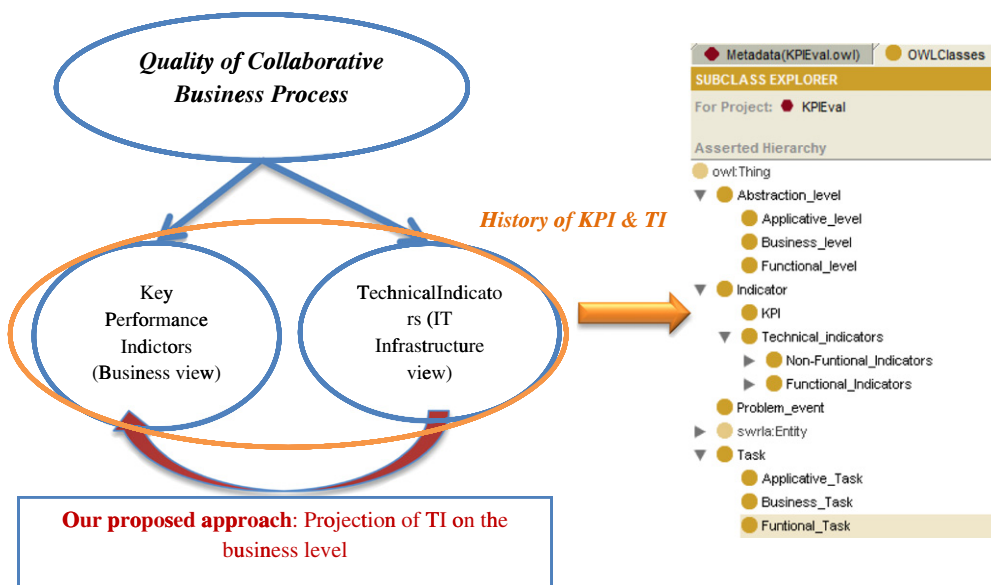


Fig. 1. Bridging the gap between the business view and the technical view of quality of BPs.

researchers calculate these two types of indicators separately. However, the business value of technical indicators has not been widely analyzed in these proposals. These existing approaches are not able to identify the gaps between business and IT activities in order to determine where the improvements might be made. They cannot align their business strategies with their IT strategies to improve their BP performance.

This work is a first step towards bridging the gap between these two views of business process quality (business level and IT infrastructure level) illustrated in Fig. 1. It is interesting to incorporate business indicators and IT indicators at the same time because they influence the overall performance of an enterprise. The performance at an applicative task is directly linked, and also has an influence and impact on the business performance. Therefore, it is highly important to link IT with business to investigate the effect of their alignment and deviation. Our proposed approach, which will be presented in the next section, is to analyze the signification of TIs at the business level using the aggregation model. The projection of these indicators on the business point of view can support organizations in achieving, assessing and maintaining alignment between business and IT.

3. Performance assessment architecture for collaborative business processes

This section presents our proposed approach for the assessment and analysis of collaborative business processes.

3.1. Collaborative business process assessment framework

As soon as the enterprise strategy has been formulated and the strategic objectives have been set, the next challenge to tackle is to adjust and maybe, even to modify the design of processes. The new design, obviously, should respond exactly to the business environment, meet the enterprise missions, collaborative objectives and interests of business actors. In fact, during the business strategy formulation, the performance objective is set up. The design of the enterprise business processes is usually modified in accordance with the enterprise strategic goals. In the framework of this strategy, companies start by identifying their business specifications and collaborative objectives in order to create a relevant model. Then, from these objectives and specifications, they define a set of KPI to evaluate their business performance. Certainly, key performance measurements help an enterprise to evaluate its readiness relative to the current and forthcoming business conditions. However, these indicators are able to give a general idea about the business appreciation. They are characterized by high level (business) interpretation and they can qualify only the collaboration between companies. In fact, companies fix the measurement frequency and calculate their KPIs. After that, the results will be interpreted and some corrections must be given to enhance the business performance. However, the measurement interval of these KPI is so long (such as week, month, year, etc.). In fact, companies have a long duration to analyze their results, to make corrections and to find solutions. They can see the impact of the solution or the correction only on the next KPI measure.

The collaborative business process is connected to a set of business processes that is a sequence of activities and actions. The concepts of enterprise architecture provide several decompositions of viewpoints. We identify in our work, three abstraction levels which are also elaborated by [35,36]:

- Business level: A simple BPMN representation targets at the business people and structuring the business process. At this level, we want to answer this question: which business tasks compose the process?
- Functional level: Further dealing the descriptive business process model with business specifications to ensure the feasibility of the process execution. We want to answer this question: how can we do it?

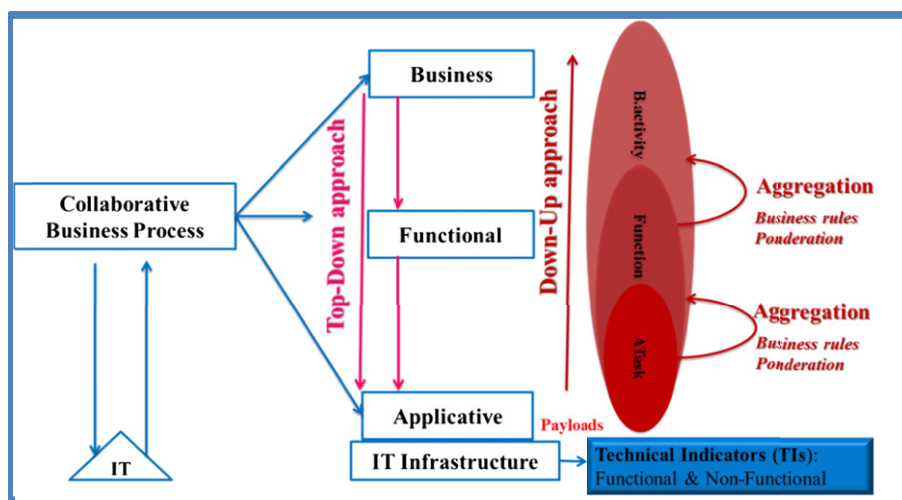


Fig. 2. The collaborative business process evaluation framework.

- Applicative level: It investigates where business processes are executed and run. Here, we can answer these questions: where do we do that? Wherewith do we do that?

We define our KPIs at the business level and our TIs at the applicative level. In fact, we limited indicators to these two levels because the functional level is created to enhance the business process specifications and to ensure the achievability of the process. It presents a transitional stage to pass from the business field to the execution environment.

We aim to evaluate the business process's performance by considering the IT infrastructures. In addition, we consider intermediate evaluations between KPI measurements. Therefore, we will zoom in on the period before KPI assessment and the period between two KPI measurements. Our CBP assessment architecture benefits from the top-down and down-up approaches. At the initial phase of development, a top-down approach is adopted. Various involved stakeholders design collaborative business processes at a very abstract level capturing the flow of business process. Then, it is enriched with the functional details of each of the tasks in the flow. Finally, the implementation details are integrated at the applicative level for the execution of business processes. Once business processes are deployed and executed, we use a down-up approach for the performance assessment of business processes. We analyze data traces generated by the execution of BPs in order to anticipate deviations and estimate problems. For this purpose, we calculate at the run-time, a set of TIs provided from IT infrastructure at the applicative level and the framework of SOA. Our main objectives are to detect the technical problem and its sources, to identify the applicative task/service that originates a problem and to correct/make some adjustments. These TIs are aggregated from the applicative level (*ATask*) to the functional level (*Function*) and then from the functional level (*Function*) to the business level (*BActivity*). In addition, TIs calculated from the monitoring of traces at real time are correlated with KPI. Finally, we compare two levels of performance: business performance calculated by KPI and performance level aggregated from TI (illustrated in Fig. 2).

Generally, we want to analyze all traces of the collaborative execution in order to anticipate problems. Therefore, we analyze the entire environment linked to the collaboration until the applicative level in the run-time and we estimate the performance of each applicative task supporting the CBP. In the next section, we will present two evaluation views (business and IT) and we will analyze the business value of IT. We have to maximize the business value of IT to ensure the alignment between business and IT.

3.2. Proposed approach for the assessment of CBP performance

We propose an evaluation method using execution traces of business processes and service statistics at the application server. Based on these values, we apply formulas to measure the key performance indicators for the high-level (business level) assessment of CBP performance.

This method presents a novel approach that focuses on the business requirements and IT supporting business process, at the same time. The main output of this approach is the consolidation levels of performance. In fact, we want to compare and correlate two performance levels: the performance level provided at the business level from KPIs and the performance level aggregated from TI. The KPIs reflect only the enterprise strategy, mission and objectives. These indicators have been identified with the ECOGRAI method [31]. The ECOGRAI method is able to drive performance, converge the objectives and assist in the decision-making process [32].

We developed our business processes into two levels (i.e., conceptual level and implementation level). At the conceptual level, various stakeholders who are involved in the CBP only know the abstract level of functionalities of each of the complex business process transactions and really do not know the execution of each of the tasks or involved transactions. Therefore, the CBP is modeled by stakeholders with the Aris Express¹ tool using BPMN 2.0 notations in order to facilitate an understanding of the performance collaborations and business transactions between the organizations and stakeholders. Then, our final model will be modeled and simulated with the business process application Activiti² by adding implementation details within each of the tasks in order to execute it. Each applicative task in Activiti is associated to a class encapsulated as a jar inside the Activiti. For the invocation of web services, which are deployed at the WSO2 application server, stub client classes are generated and used.

Once we have our business processes deployed successfully, the Activiti Process Execution Engine automatically captures data traces that happen during the BP execution in the Activiti databases.

We create a reference analysis framework (Table 1) that exploits the data collected from the business process execution environment (SOA/BPM) given from the IT infrastructure at the run time. The execution environment (SOA/BPM) (illustrated in Fig. 3) provides the structured data in the different databases (SOA engine, ERP, etc.) in order to facilitate the exploitation of the process tracking data and related payloads.

At the applicative side, it is useful to measure and evaluate the quality of deployed processes. At this level, we characterize applicative tasks through two sets of indicators, as discussed in Table 1.

- Functional indicators: related to the running environment of each service task. It is about quantitative indicators characterizing data input/output, assigned organization role, the implementation type of the task (i.e., service task, user task, etc.) and duration. These indicators (implementation type, input/output and role) are used and don't have an impact on the performance.
- Non-functional indicators: related to the run-time aspects of BP such as maturity, availability, risk level, frequency, etc.

¹ <http://www.ariscommunity.com/aris-express>.

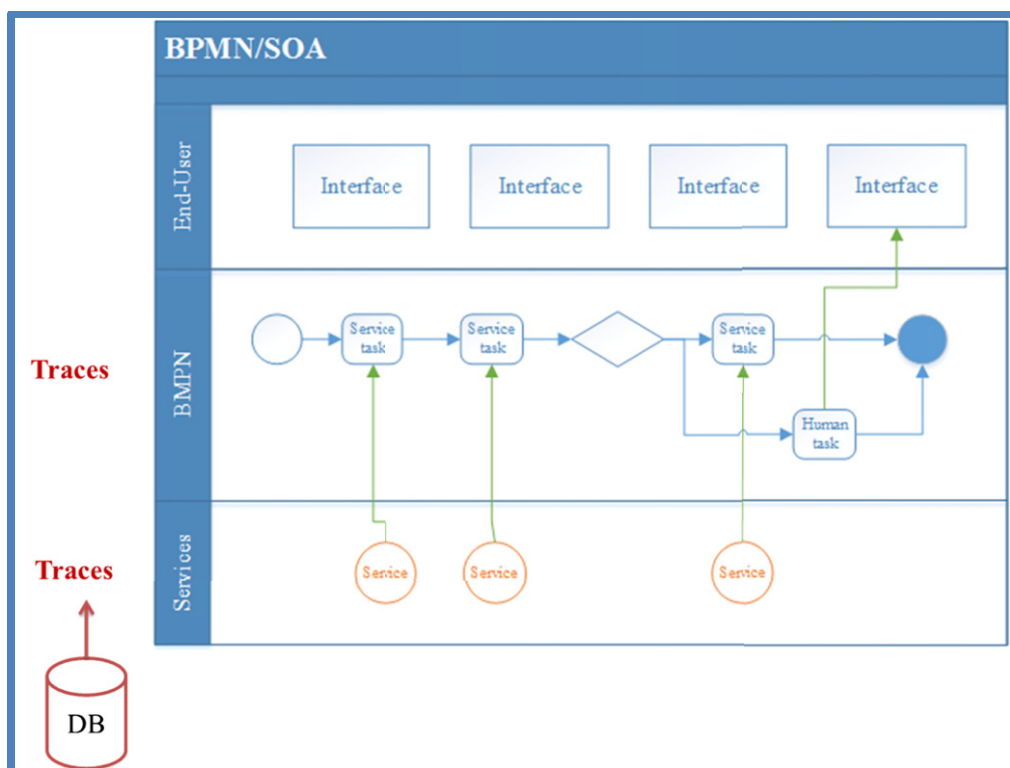
² <http://activiti.org/>.

Table 1

The reference analysis framework.

Technical indicators (TI)	Concept details	
Functional	Implementation type	User task: needs to be done by a human actor Manual task: is external to the BPM engine, it passes through Activiti Service task: is used to synchronously invoke an external Web service
	Input Output Role Execution duration	Number of parameters Number of parameters Internal actor/external actor Time between start execution time and end execution time of the task Execution duration/average execution duration of the same type of this task
Non-functional	Status	Task completed (100%) or uncompleted (0%)
	Maturity	Level of maturity: (initial, managed, defined, quantitatively managed and optimizing)
	Risk	% risk for success = probability compared to status (completed or not) * gravity Gravity: sub-process 1→8 Sub-process 2→7 Sub-process 8→1
	Frequency Availability Fault count Interoperability % of applicative task performance	Number of calls Number of successful answers Number of faults while invocations of service task Expected exchanged data/total number of exchanged data

These non-functional indicators are calculated from the data traces generated in the Activiti and WSO2 server. As our web services are deployed at the WSO2 application server; therefore, we explore its databases for the calculation of non-functional indicators, such as frequency and availability. For an example, Fig. 4 illustrates the statistics of a service named *AuthenticateUser*

**Fig. 3.** Business process performance in BPM/SOA-based platform.

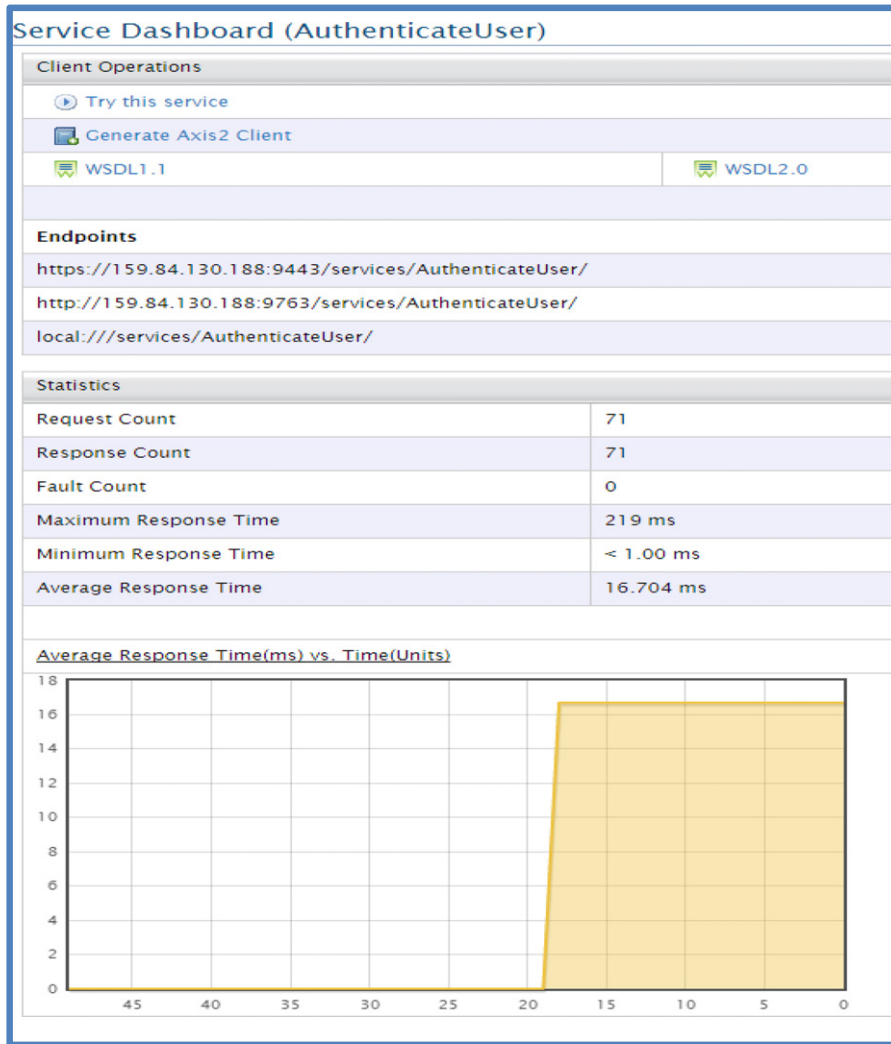


Fig. 4. Statistics of a service from WSO2 application server.

deployed at the WSO2 server which are used to calculate these indicators. We identify, for each non-functional indicator, its concepts and dimensions and how it can be measured in our proposed approach:

- Status—It is the state of the task, whether it is successfully completed or not.
- Maturity—It is a structured collection of elements that describe the characteristics of effective processes at different stages of the development [52]. A maturity model consists of a sequence of maturity levels for a class of objects. It represents an anticipated, desired, or typical evolution path of these objects shaped as discrete stages [53]. Table 2 discusses various levels of maturity and along with their potential quantification. Our CBPs are assessed using the CMMI model: initial, repeatable, defined, managed and optimized. For each level, we associate the appropriate ponderation in order to facilitate the calculation of CBP maturity.

Table 2
Maturity levels [34].

Maturity level	Explanation	Potentiality quantification
Level 1: Initial	No reliable process, no control, general indiscipline	20%
Level 2: Managed	Disciplined process	40%
Level 3: Defined	Standardized, improved processes	60%
Level 4: Quantitatively managed	Quantified, systematic application of measurement processes	80%
Level 5: optimizing	Continuous improvement, control of change	100%

- Risk—It is the chance of something happening during the business process execution that will have a negative impact on the process objectives, and it can be measured in terms of consequences. Our CBP is composed of complex business process transactions performed by many stakeholders. Therefore, we partitioned it into eight sub-processes, each running in a separate pool, performing a set of activities by an individual actor. Once one sub-process finishes its functionality, it invokes the next sub-processes to start. Therefore, gravity is the number of all sub-processes involved in the CBP. It also means that the risk is related to the gravity of CBP, i.e., risk of CBP is the aggregation of risk of each of the sub-processes.
- Frequency—It is the number of calls performed for the service task in order to perform the desired task. The parameter *request count* in the service statistics helps to measure the desired value.
- Availability—Service task can be considered available when it is simply expected to be delivered reliably and on demand at the desired moment. Unavailable services can disturb the execution of business processes causing a bad impact. We calculate this value by dividing *response count* by the *request count* from the service statistics.
- Fault Count—It is a count of a service when it has received an invocation, but has not responded due to some reason. We analyze the value of *fault count* from service statistics for this indicator.
- Interoperability—It refers to the ability of two or more systems or components to exchange information and to use the information that has been exchanged [54]. Enterprise interoperability is defined as the ability of an enterprise to collaborate with its business partners and to efficiently establish, conduct and develop information technology (IT)-supported business relationships with the overall objective of creating value [55]. In business process application based on SOA, services play an important role for enabling interoperability for establishing collaborations between organizations [48]. It is the capability to exchange data (messages, information, etc.) between stakeholders of organizations involved in the collaborative business processes.

For the aggregation at the applicative process level, we apply a serialization mechanism for the process tasks in order to solve parallel branches by the maximum value rule. After that, we assume the following composition rules:

- The applicative level is composed of a set of applicative tasks
- The functional level is composed of a set of functions
- The business level is composed of a set of business activities
- Each applicative task is a sub-class of a functional task



Fig. 5. Ontological model (classes).

- Each functional task is a sub-class of a business task
- No aggregation between functional indicators and non-functional indicators to make sense of our capitalization

The passage from one level to higher level is achieved using weights and aggregation rules (min, max, average, etc.). We emphasize that there is no weighting between functional and non-functional indicators. Therefore, there is no regrouping of functional and non-functional indicators in the final aggregation at the business level. The aggregation model is able to evaluate CBP from the execution traces and to anticipate deviations and events that block business performance.

In addition, we need to collect and structure the performance knowledge, the measurements and the related analysis in order to correlate it with the frequency dimension and predict business performance degradation. Otherwise, in the conceptual viewpoint, the structure of our analysis model evolves during the time. Besides, the process environment is dynamic and uncertain. Consequently, the interpretation of results, the relationships between concepts and the correlation between events will be incremental during our analysis and assessment. An ontology is capable of analyzing and assessing BP performance taking into account the evolution of company events because it is characterized by its scalability. In addition, an ontological model enriches the evaluation of BP models, capitalizes on BP assessments and analyzes tendencies for the next analysis.

Therefore, we have designed our ontology in Protégé 3.5 (illustrated in Fig. 5) for the conceptualization of our domain knowledge to support the aggregation model in the three abstraction levels (business, functional, applicative). We analyzed the consistency of our ontological model with PELLET³ description logic reasoner. In this model, we integrated execution results, its analysis history and decisions taken in case of problems or events. We use ontology in order to compare and always link the new configuration or the new measured values given from a new execution with past scenarios and resulting decisions that have been registered in the database of our ontology.

In our ontological model, we created four principal concepts and properties (illustrated in Fig. 6) that are connected together and present a support to take the best decision on the performance:

- Abstraction levels: decompositions of business process viewpoints
- CBP concepts: set of all indicators related to business and IT supporting business process
- Problem/event: can be defined as any detectable or discernible occurrence that has significance for the management of the IT infrastructure or the delivery of IT services
- Task: decomposition of business process

Our ontological model has the following properties to capture the values of CBP:

- Fig. 6a illustrates a list of data type properties also called *Attributes*, which contain additional data information about those concepts, for example, the 'start time' of a task.
- Fig. 6b illustrates list of Object type properties also called *Relationships* which express how two concepts relate to each other, such as KPI 'calculated at' business level.

The important task of populating this ontological model with instances is our next step. We have used Activiti tool for the execution of our collaborative business processes. During the execution of business processes, Activiti tool generates data traces in an SQL Server database. Based on the data mining tool, we dig out data traces from the Activiti history to populate our ontology with instances, for example, the service task "authenticate" is completed at a duration of 253 ms. Fig. 7 illustrates the instance tab to depict various instances of applicative task. Once our ontological model has execution traces, our SWRL-based semantic rules calculate the values of indicators such as availability, risk, priority, etc., as represented by Fig. 9. Aggregation rules from applicative tasks to the business tasks help in order to find a qualification of the performance at the business level. Then, based on SWRL rules which are illustrated in Fig. 8, our semantic execution engine detects the faulty tasks and generates alarms for the enterprise. Detection of faulty tasks and alarms are helpful for making further queries and to take the best decision in the company.

We present above the structuring concepts in our ontological model in order to apply the aggregation rules. In the following section, we explain the exploitation of the knowledge coming from traces in order to make the best decision in the company.

3.3. Exploitation model

To better explore the huge volume of data at the run-time and to ensure the robustness of our solution, the traces provided from Activiti databases (DB) have to be integrated in a repository that is extern of the ontological model. We consider that our ontology should contain just the reasoning aspects. For that, we propose retrieval mechanisms (illustrated in Fig. 10) to fetch data from the real system and to process these data until our ontological model. It is not a direct exploitation of data in the Activiti DB because we need some persistence intermediate DB for computation and treatment of data traces. There are some transformations and some loading problems for different individuals with data coming from traces that have to be done externally, outside of the ontology. In this case, the ontology merely defines the tasks (applicative, functional and business) with some attributes and some values and it contains the aggregation rules and the transformations (illustrated in Fig. 9) in order to be more robust and to reduce the large volume of data in its repository.

Therefore, the instances of Activiti should be transformed automatically, through some SQL queries and scripts, in a template that contains the reference analysis framework defined above in Table 1. This template is considered as a calculation model and a

³ <http://clarkparsia.com/pellet/>.

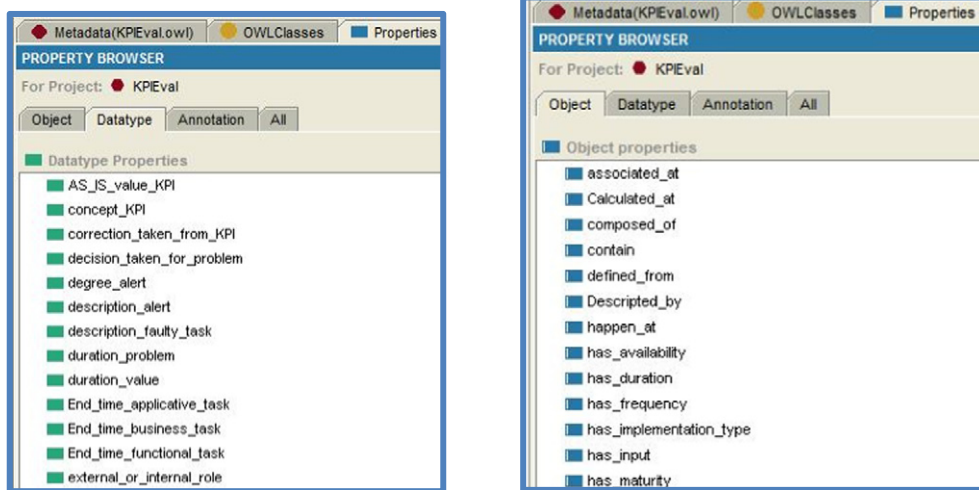


Fig. 6. Properties in our ontological model. (a) Data type properties and (b) object type properties.

preparation treatment of data to be inserted inside the ontology. Through the application of this template, we introduce in the ontology just adequate information of the Activiti task using intermediate transformations based on some transformation and aggregation rules. In addition, the ontological model contains the history of results for some definite periods in order to take decision in the company and to identify if there is some noise or other problems. The frequency of upload traces coming from Activiti DB in the template and the frequency to query the real system have to be fixed in order to supervise the aggregation performance and the performance at the business level.

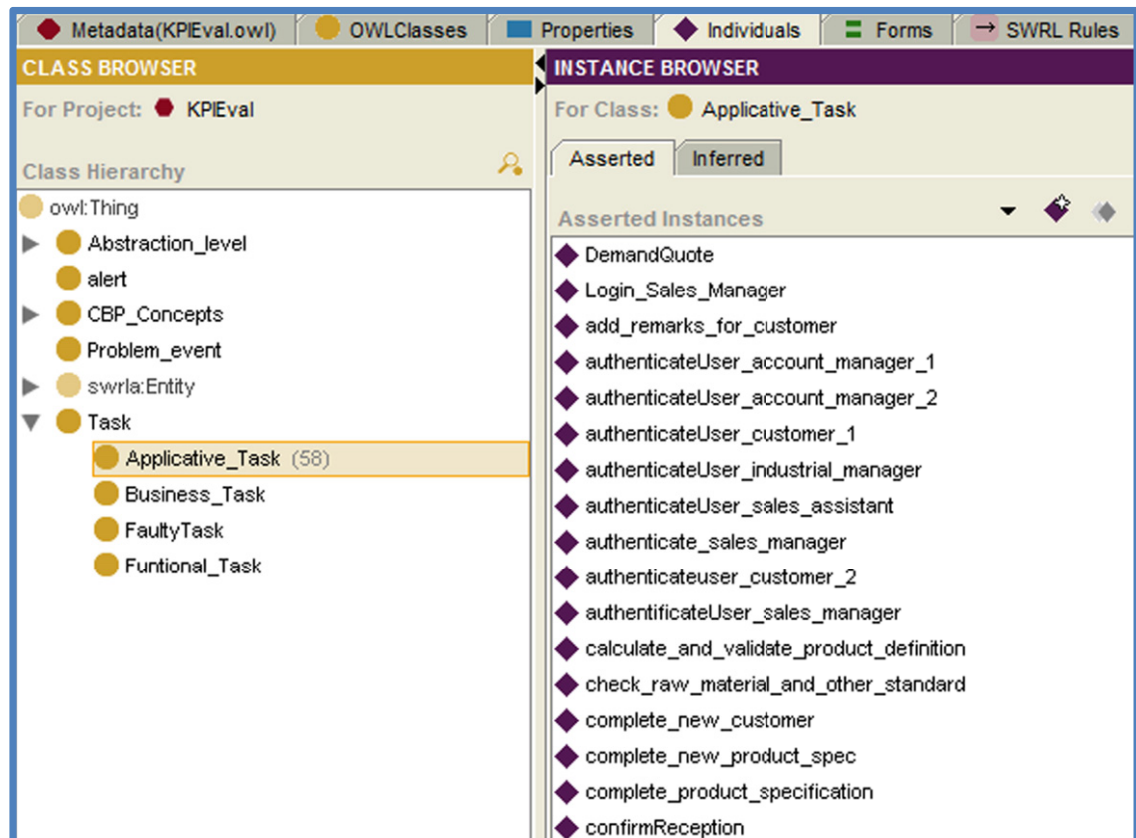
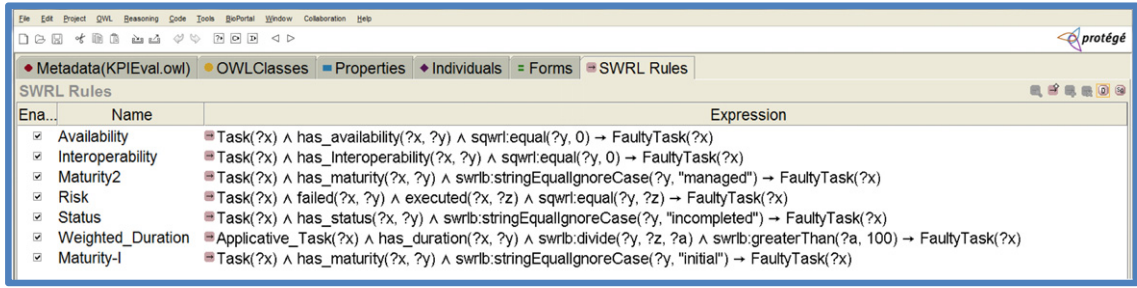


Fig. 7. Example of individualization in our ontological model.



Ena...	Name	Expression
☒	Availability	$\text{Task}(?x) \wedge \text{has_availability}(?x, ?y) \wedge \text{sqwrl:equal}(?y, 0) \rightarrow \text{FaultyTask}(?x)$
☒	Interoperability	$\text{Task}(?x) \wedge \text{has_interoperability}(?x, ?y) \wedge \text{sqwrl:equal}(?y, 0) \rightarrow \text{FaultyTask}(?x)$
☒	Maturity2	$\text{Task}(?x) \wedge \text{has_maturity}(?x, ?y) \wedge \text{swrl:stringEqualIgnoreCase}(?y, \text{"managed"}) \rightarrow \text{FaultyTask}(?x)$
☒	Risk	$\text{Task}(?x) \wedge \text{failed}(?x, ?y) \wedge \text{executed}(?x, ?z) \wedge \text{sqwrl:equal}(?y, ?z) \rightarrow \text{FaultyTask}(?x)$
☒	Status	$\text{Task}(?x) \wedge \text{has_status}(?x, ?y) \wedge \text{swrl:stringEqualIgnoreCase}(?y, \text{"incompleted"}) \rightarrow \text{FaultyTask}(?x)$
☒	Weighted_Duration	$\text{Applicative_Task}(?x) \wedge \text{has_duration}(?x, ?y) \wedge \text{swrl:divide}(?y, ?z, ?a) \wedge \text{swrl:greaterThan}(?a, 100) \rightarrow \text{FaultyTask}(?x)$
☒	Maturity-1	$\text{Task}(?x) \wedge \text{has_maturity}(?x, ?y) \wedge \text{swrl:stringEqualIgnoreCase}(?y, \text{"initial"}) \rightarrow \text{FaultyTask}(?x)$

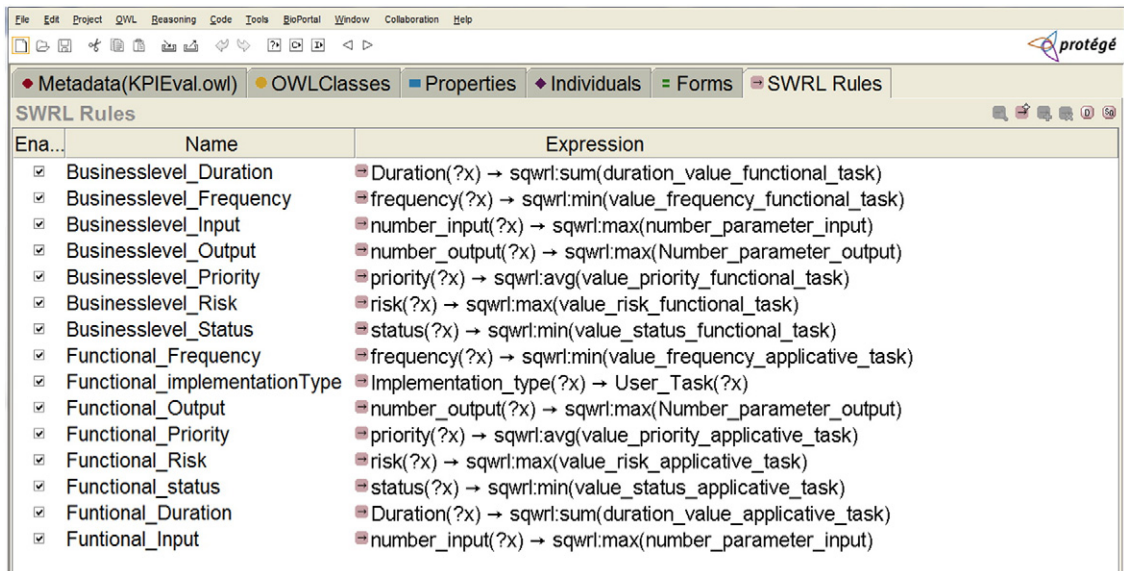
Fig. 8. Rules for taking decision.

The most relevant aspect in our approach is not to obtain and calculate indicator values or the aggregation value from IT. However, we are interested in tracking these values that we are evaluating at the abstraction levels in some periods. We analyze the behaviors of BPs and we focus on the evolution of the different measurements and the knowledge obtained from the executions during the run time. Then, the decision made is based on the knowledge obtained from our measurements and the history of results of some executions in a precise period. By the ontology, we save the performance trajectory of the business process in order to predict all situations which can cause noise and malfunctioning.

In addition, our ontological model helps to reuse the information in the future and to synchronize with other kinds of real decomposition and analysis of business processes in reality. In the applicative point of view, this model is able to structure our approach; otherwise, the CBP is always connected to several BP that are defined in some quality systems of the company. Besides, these BPs are described at the high levels of management and in these quality systems which define several business indicators.

This ontology seeks not only to generate value but also to be open to any interaction with others models related to the existing BPs in the company and to each quality management system that define the KPIs. Therefore, our aim, through this ontological model, is to maximize the history of data at the different abstraction levels in order to merge it with other reference models, such as the SCOR model (supply-chain operations reference), which treats other real decompositions and structures of business process related to the functional aspects (process, process details and procedure). In our perspective, we want to connect and correlate our down-up approach (from IT level to business level) with other business process management approaches to enrich our model with more specificity and events. For that, our aggregation rules should be adapted in order to adjust the reactivity of the system in case of noise or problems, taking into account the other structuring of BPs in the company. For example, the SCOR model deals with an ontology [37]. We will get from this ontology some concepts that help to select, adapt and change the aggregation rules placed on our ontological model.

The proposed exploitation model is used to explore the huge volume of information provided from the execution of the collaborative business process. Therefore, we are developing this model and we are testing the connection between its different components (ontology, intermediate DB, etc.). In our future work, we aim to evaluate the sensitivity and the relevance of our model. In this way, we will use the traces of six different collaborative business processes to validate the strictness of our proposed approach.



Ena...	Name	Expression
☒	Businesslevel_Duration	$\text{Duration}(?x) \rightarrow \text{sqwrl:sum}(\text{duration_value_functional_task})$
☒	Businesslevel_Frequency	$\text{frequency}(?x) \rightarrow \text{sqwrl:min}(\text{value_frequency_functional_task})$
☒	Businesslevel_Input	$\text{number_input}(?x) \rightarrow \text{sqwrl:max}(\text{number_parameter_input})$
☒	Businesslevel_Output	$\text{number_output}(?x) \rightarrow \text{sqwrl:max}(\text{Number_parameter_output})$
☒	Businesslevel_Priority	$\text{priority}(?x) \rightarrow \text{sqwrl:avg}(\text{value_priority_functional_task})$
☒	Businesslevel_Risk	$\text{risk}(?x) \rightarrow \text{sqwrl:max}(\text{value_risk_functional_task})$
☒	Businesslevel_Status	$\text{status}(?x) \rightarrow \text{sqwrl:min}(\text{value_status_functional_task})$
☒	Functional_Frequency	$\text{frequency}(?x) \rightarrow \text{sqwrl:min}(\text{value_frequency_applicative_task})$
☒	Functional_implementationType	$\text{Implementation_type}(?x) \rightarrow \text{User_Task}(?x)$
☒	Functional_Output	$\text{number_output}(?x) \rightarrow \text{sqwrl:max}(\text{Number_parameter_output})$
☒	Functional_Priority	$\text{priority}(?x) \rightarrow \text{sqwrl:avg}(\text{value_priority_applicative_task})$
☒	Functional_Risk	$\text{risk}(?x) \rightarrow \text{sqwrl:max}(\text{value_risk_applicative_task})$
☒	Functional_status	$\text{status}(?x) \rightarrow \text{sqwrl:min}(\text{value_status_applicative_task})$
☒	Functional_Duration	$\text{Duration}(?x) \rightarrow \text{sqwrl:sum}(\text{duration_value_applicative_task})$
☒	Functional_Input	$\text{number_input}(?x) \rightarrow \text{sqwrl:max}(\text{number_parameter_input})$

Fig. 9. Aggregation rules from applicative tasks to business tasks.

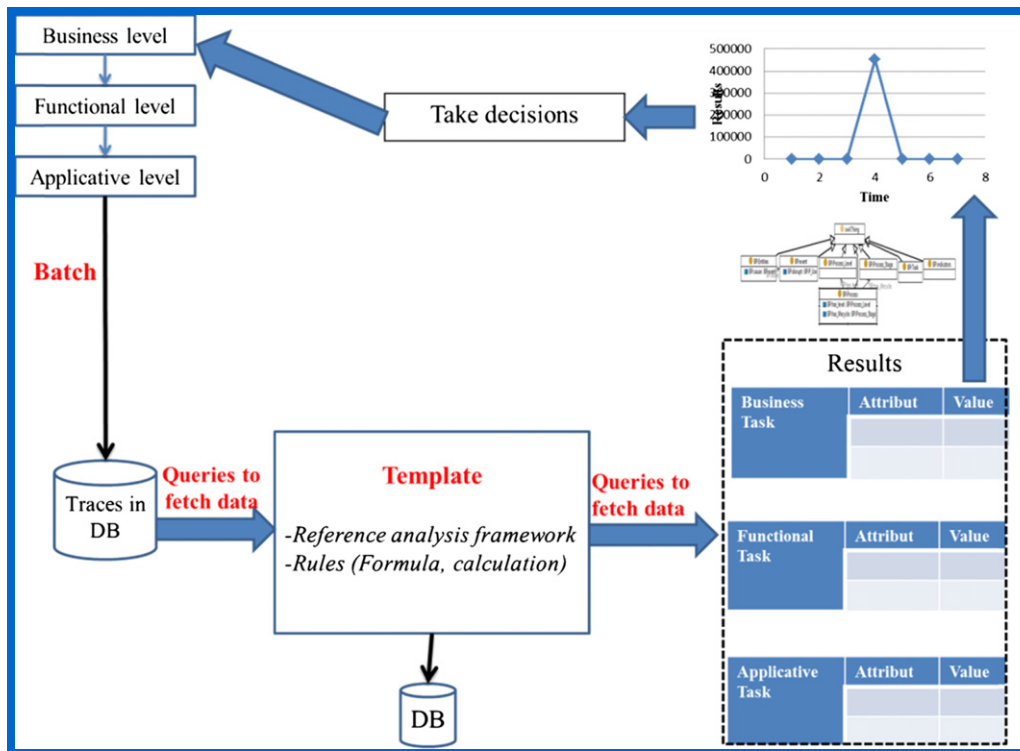


Fig. 10. The exploitation of execution traces.

4. Results and interpretation

4.1. Case study

In the next section, we present our execution results of this collaborative business process and its interpretations.

We study a collaborative business process from APR (Applications Plastiques du Rhône) company. The business process, which is named *create quote*, focuses on the management of requests for price. By quote structuring and automation, APR aims to enhance the collaboration infrastructure with customers in order to enhance communication flows when initiating collaborations. The main objective targets to accelerate the answer of quotes for the customer projects. When the customer's project is uploaded, the sales assistant is notified. She/he checks and validates the customer information (name, contact, guarantee amount paid, etc.). After that, a notification will be sent to the account manager. He validates the product and checks the raw material needed. The purchase department consults supplier to take ideas about the price and the delay of the raw materials and needs. Then, when this department receives answers from suppliers, it sends all these answers to the industrial manager to complete the product information and to validate the quote. The final quote containing the price will be communicated with the customer. The value-added contribution of the process "create quote" is a new concept of how to meet the customer's needs and calculate the price. Therefore, in our project FITMAN, we targeted two KPIs (discussed in Table 3), which are defined with the ECOGRAI method based on the business process objectives. First, KPI targets to improve the quality of the quote creation and response to the customer. Second, KPI targets to examine the number of unsuccessful quotes demanded by the clients due to their high prices. When the value of these indicators is higher than the fixed/target value by the APR, these indicators investigate the cause which may happen at the business level or potentially at

Table 3

Key performance indicators for "create quote".

CBP name	Create Quote	
Strategic objective	Improve the quality of customer relationships	
KPIs	KPI 1	KPI 2
KPI definition	Time limit for responding of quote demands (old/new product)	Percentage of the unsuccessful quotes due to the high price
Tactical objectives	Improve the quality of quote	Improve the competitiveness of customer pricing
Operational objectives	Improve the delay of answer to the quote	Control the cost of a quote
Formula	Time between the date of response to the customer and the date of customer quote request	Number of unsuccessful quotes due to high price * 100/total number of quotes processed

applicative level. The abnormal value of these KPIs demonstrates the degradation of TIs (such as un-availability of services, no response of applicative task, faulty service occurrence, percentage risk of BP success, incomplete human task, etc.)

The calculation of KPIs depends on the flow and level of different parts during the execution of BPs:

- KPI 1: Totality of the CBP from the demand of the customer to the final quote containing the price.
- KPI 2: Part of validate product specifications and part of complete product specifications for the analysis of product specifications and calculate the price.

On the one hand, when a problem has been detected at the highest level (business level) in the flow, we can zoom in on the applicative level in order to know the technical problem. Therefore, we focus on the applicative tasks that belong to this flow.

We configured Activiti with MS SQL Server for the persistence storage of BP execution history so that we can exploit data traces for the assessment of our CBP at runtime. Fig. 11 illustrates the Activiti history details of our BP executions in the SQL Server.

After every thirty minutes, we executed our CBP consecutively four times. Then, we tracked the evolution of this CBP in two hours. We analyzed that in the three executions, the aggregation values of technical indicators (duration, maturity, risk, etc.) are approximately close to each other. But, in the fourth execution, we observed that the customer had a problem with uploading quoted project files. When he connected to the collaborative platform, he did not directly integrate his files related to his quote because he lost a long time collecting his documents. The collaborative business process stopped running because this technical problem appeared. The service task, *register demand*, did not answer due to the time-out occurrence. To overcome this type of problem, the company should ask their customer(s) to prepare all the documents before connecting to the system to optimize the duration of the upload project. Table 4 shows the execution results of the aggregation model at the business level. For example, the technical indicator risk, with an average value of 37% based on three executions, shows an early warning to the potential events that may harm the continuity of the CBP execution.

4.2. Evaluation and interpretation of our approach

Through the application of the proposed methodology, we are proposing performance architecture interoperated with the legacy system from where we collect IT traces in the BPM-SOA-based environment. This methodology allows us to identify the source of performance deviation events and predicts it in the future in order to be more reactive. Besides, it helps to analyze deviation characterizations (duration, frequency, level, source, impact, etc.) and the performance trajectory of business process regarding the business performance level.

***** Script de la commande SelectTopRows à partir de SSMS *****

```

SELECT TOP 1000 [ID], [TYPE], [PROC_INST_ID], [EXECUTION_ID], [TASK_ID], [ACT_INST_ID],
[NAME], [VAR_TYPE], [REV], [TIME], [BYTEARRAY_ID], [DOUBL], [LONG], [TEXT], [TEXT2]
FROM [Fahad].[dbo].[ACT_HI_DETAIL]

```

ID	TYPE	PROC_INST_ID	EXECUTION_ID	TASK_ID	ACT_INST_ID	NAME	VAR_TYPE	REV	TIME	BYTEARRAY_ID	DOUBL	LONG	TEXT	TEXT2
1	VariableUpdate	2507	2507	NULL	NULL	initiator	string	0	2015-02-25 16:47:00.880	NULL	NULL	NULL	kermil	
2	FormProperty	2507	2507	2513	2512	Name	NULL	NULL	2015-02-25 16:47:16.177	NULL	NULL	NULL	marous	
3	FormProperty	2507	2507	2513	2512	APR_id	NULL	NULL	2015-02-25 16:47:16.177	NULL	NULL	NULL	123	
4	FormProperty	2507	2507	2513	2512	Role	NULL	NULL	2015-02-25 16:47:16.177	NULL	NULL	NULL	NULL	
5	VariableUpdate	2507	2507	NULL	2512	Name	string	0	2015-02-25 16:47:16.187	NULL	NULL	NULL	marous	
6	VariableUpdate	2507	2507	NULL	2512	APR_id	string	0	2015-02-25 16:47:16.187	NULL	NULL	NULL	123	
7	VariableUpdate	2507	2507	NULL	2512	Role	null	0	2015-02-25 16:47:16.187	NULL	NULL	NULL	NULL	
8	VariableUpdate	2507	2507	NULL	2523	status	string	0	2015-02-25 16:47:16.197	NULL	NULL	NULL	ok	
9	VariableUpdate	5001	5001	NULL	NULL	initiator	string	0	2015-02-25 17:11:09.783	NULL	NULL	NULL	kermil	
10	FormProperty	5001	5001	5007	5006	Name	NULL	NULL	2015-02-25 17:11:24.780	NULL	NULL	NULL	fahad	
11	FormProperty	5001	5001	5007	5006	APR_id	NULL	NULL	2015-02-25 17:11:24.783	NULL	NULL	NULL	fahad	
12	FormProperty	5001	5001	5007	5006	Role	NULL	NULL	2015-02-25 17:11:24.783	NULL	NULL	NULL	fahad	
13	VariableUpdate	5001	5001	NULL	5006	Name	string	0	2015-02-25 17:11:24.783	NULL	NULL	NULL	fahad	
14	VariableUpdate	5001	5001	NULL	5006	APR_id	string	0	2015-02-25 17:11:24.783	NULL	NULL	NULL	fahad	
15	VariableUpdate	5001	5001	NULL	5006	Role	null	0	2015-02-25 17:11:24.783	NULL	NULL	NULL	NULL	
16	VariableUpdate	5001	5001	NULL	5017	status	string	0	2015-02-25 17:11:24.810	NULL	NULL	NULL	ok	

Exécution de requête réussie. FAHAD_PC (11.0 RTM) Fahad_PC\Fahad (S2) master 00:00:00 16 lignes

Fig. 11. Activiti history details in SQL Server.

Table 4

Execution results of aggregation model at the business level.

CBP: create quote	Indicators	Execution 1	Execution 2	Execution 3	Execution 4
Functional	Implementation type	User task	User task	User task	Not completed because of a technical problem
	Input	6	6	6	
	Output	6	6	6	
	Role	Extern	Extern	Extern	
	Duration	452506,5	513012	445965,5	
	Priority	50%	50%	50%	
Non-functional	Status	100%	100%	100%	
	Maturity	60%	60%	60%	
	Risk	35,96	40,78	34,98	
	Frequency	100%	100%	100%	
	Availability	100%	100%	100%	
	Interoperability	24,56%	23,98	23,78	

Our ontological model-based knowledge repository is able to collect and structure the semantics of the performance deviation events. In fact, this model integrates the history of business process analysis, the detected events, the deviation scenarios and the decision making in the three abstraction levels (business, functional and applicative). It contains the performance knowledge (execution and analysis) in order to correlate it with the high-level frequency and predict business performance deviation.

Finally, in the framework of our approach, we implement a decision model which is based on the history of analysis and IT traces. This decision model quickly detects the business collaboration degradation. This decision model focuses on the business rules defined in our ontological model.

The first results obtained from our methodology are encouraging and significant. Therefore, the experiments will be conducted on other different collaborative business processes in order to enhance our outcomes in the future.

The first results show that our proposed approach presents some limitations. In the first hand, there is the problem of abstraction levels in the aggregation model. In fact, we can mention the causality of problems between the three levels (business, functional and applicative). For example, when we aggregate the technical indicators, some problems may be hidden by other problems. Besides, the use of the compensation (average, weight, etc.) has to be justified. In addition, the correlation between the assessment level and the business decision is considered a complex task.

On the other hand, there exists the problem of the performance dimensions. Indeed, different levels of granularity can be considered for measuring the quality in an organization. Evaluating quality at different levels of granularity can involve a wide range of issues, from measuring the achievement of an organization vision at the highest level to measuring the quality of services of a business process.

5. Conclusion and future works

The information provided from the execution of the collaborative business process constitutes the main elements which are used for the better understanding and communication of an enterprise's processes and compares the functioning of the organization with its business goals. The measurement of their business process evaluation is a key issue to be able to analyze their operations to investigate if business objectives are fulfilled. It is therefore important to track information during the business process execution at each dimension (business, IT, etc.) in order to evaluate the performance of the collaborative business processes and propose some improvements. This paper deals with an essential issue in this context, namely, the analysis and assessment of collaborative business processes. We presented *performance assessment architecture* for the collaborative business processes in the BPM-SOA-based environments. To address the discussed issues, our approach is based on the execution traces of business processes combined with the assessment method using the key performance indicators. On one hand, we defined from the business objectives, using ECOGRAI method, a set of KPIs to evaluate the business performance of the enterprise. On the other hand, we took into account IT infrastructures supporting these CBPs. For that, we analyzed traces from the data resulting from IT in order to anticipate problems and deviations, and we calculated from these traces some technical indicators (risk, interoperability, maturity, etc.) at the applicative level (execution level). The technical indicators are aggregated from the applicative level to the functional level and from the functional level to the business level using the semantic-based aggregation rules. The aggregation model (based on our down-up approach) is able to evaluate CBP from the execution traces and to anticipate deviations and events that block business performance. In order to collect and structure the semantics of business process performance, we developed an ontological model-based knowledge repository. This model is able to store the performance trajectory of business processes in order to predict possible situations providing different noise and problem. Its objective is to maximize the history of data at the different abstraction levels in order to merge it with other reference models that deal with other real decompositions and structures of business processes. The ontological models supporting our aggregation model is validated regarding the consistency of the concepts and their relations. The absence of inconsistencies was a necessary formal condition for the completion of the ontology development process. The precise track of execution data in our approach helps to reflect the actual functioning of the BP and to know possible events disrupt the proper functioning of BPs. We conclude that our first results are significant and they can target the sources of dysfunction in the process. However, these results also show that our proposed approach presents some limitations. For example, it doesn't take into account the problem of abstraction levels in the aggregation model.

In our future work, we will focus on the improvement and the relevance of our proposed approach. Therefore, the correlation problems in the aggregation model will be analyzed. Then, we will adapt and adjust our aggregation rules in order to be more reactive for the

future problems or deviations. In addition, we will connect and correlate our down-up approach with other business process management approaches to enrich our model with more specificity and events. Then, we will test other approaches using our CBPs in order to compare our results and to concentrate on the pertinence, the sensitivity of the business changes and the capacity of decision-making.

Acknowledgment

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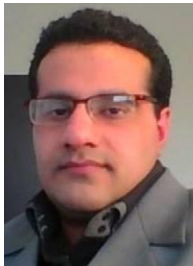
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